



Topics for Today

0. Linear Shift Invariance: Recap

I. Impulse Response

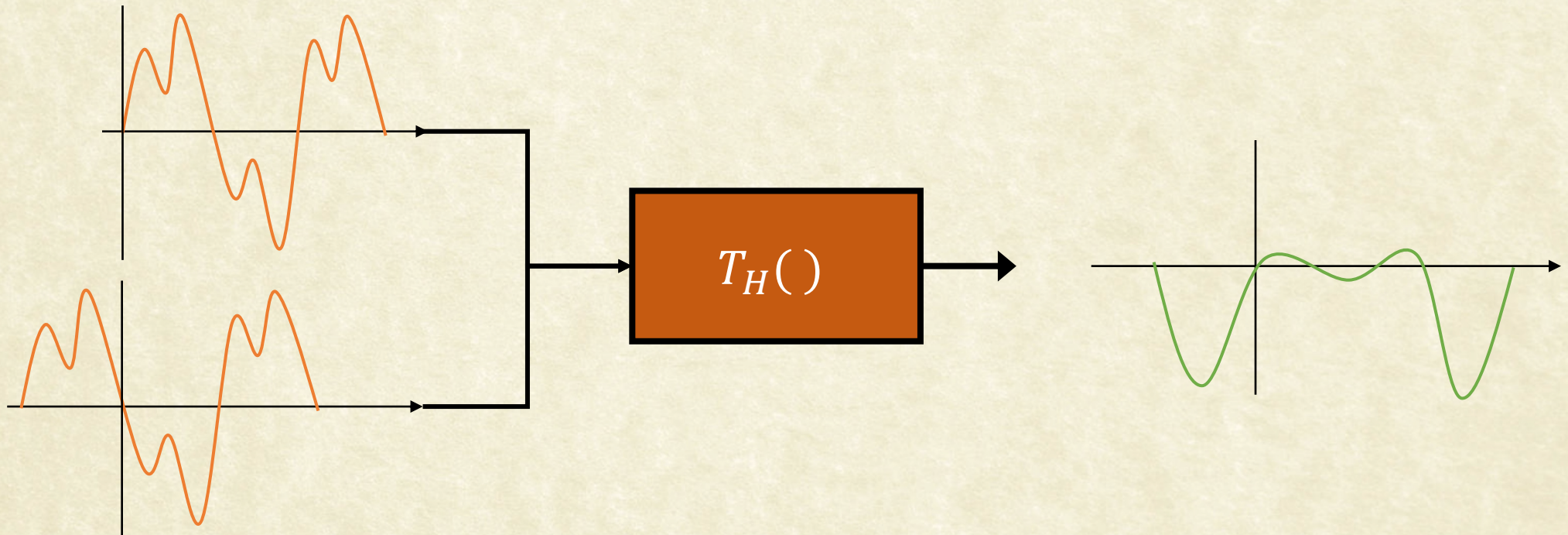
II. Separability

III. Frequency Domain Basics



CS7.404: Digital Image Processing

Monsoon 2025: **0**: Linear Shift Invariance

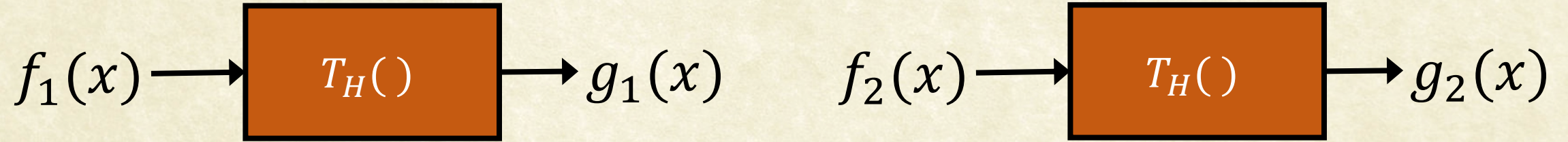


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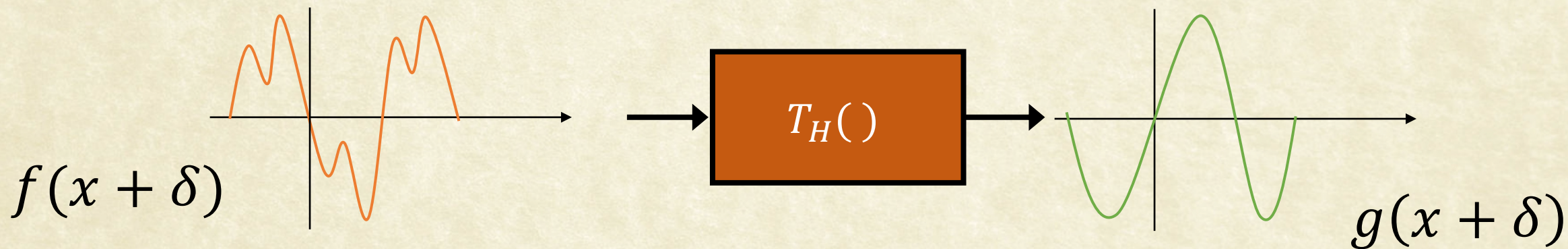
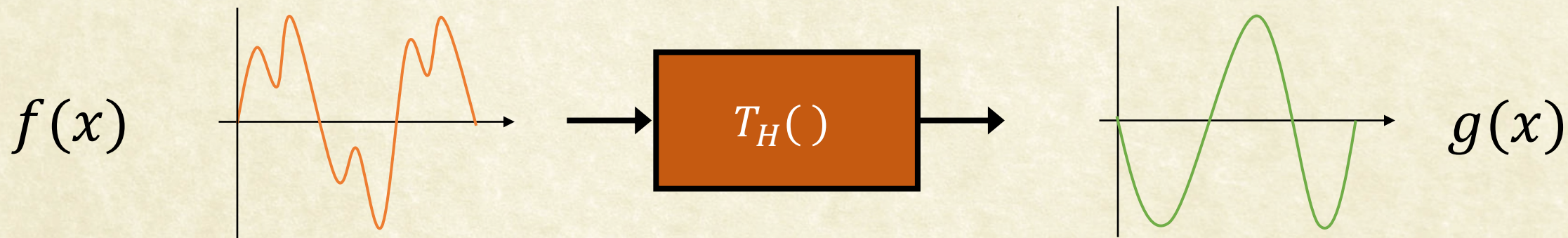
Linear Systems

$$g(x) = T_H(f(x))$$





Shift Invariant Systems





Linear Shift Invariant Systems

- Systems that are both Linear and Shift Invariant
- All convolution operations are Linear Shift Invariant
- All Linear Shift Invariant Systems can be expressed as Convolutions
- Characterized by Impulse Response or PSF
- Additional Properties
 - Separability

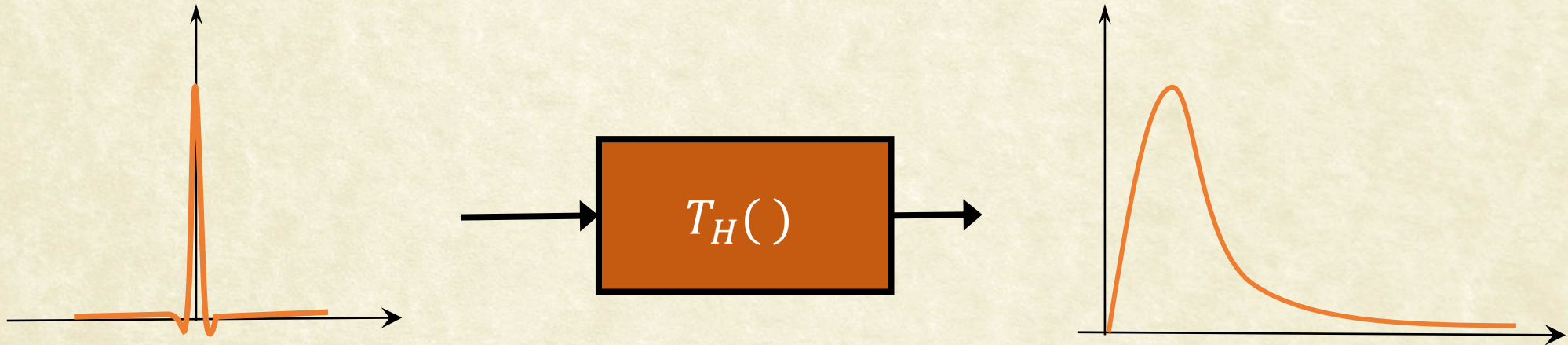


Questions?



CS7.404: Digital Image Processing

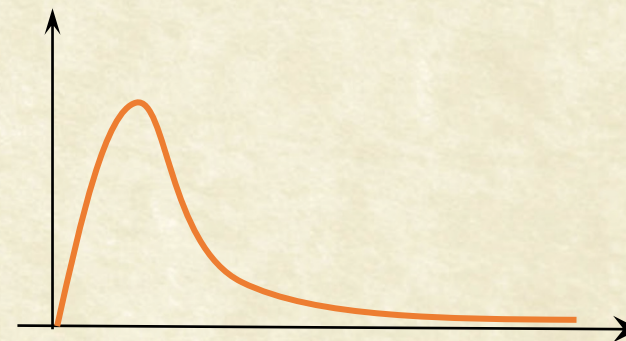
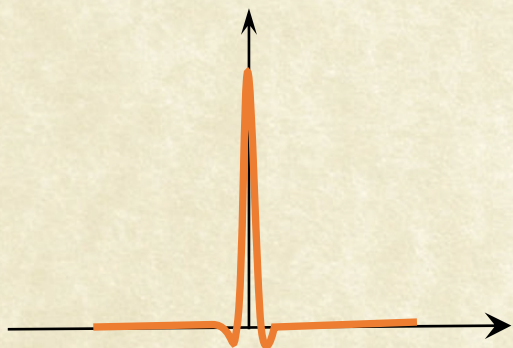
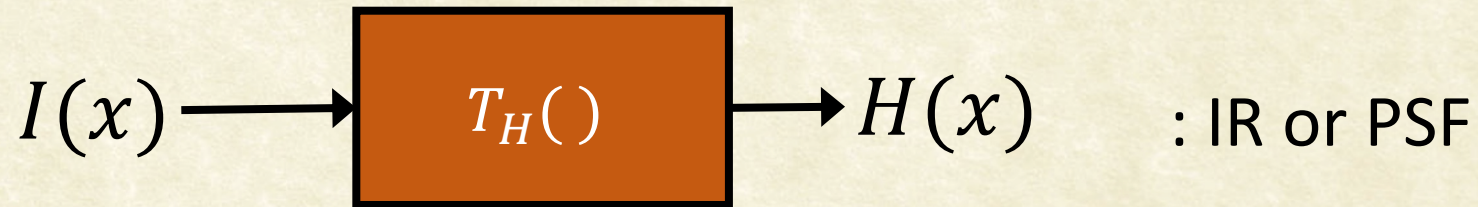
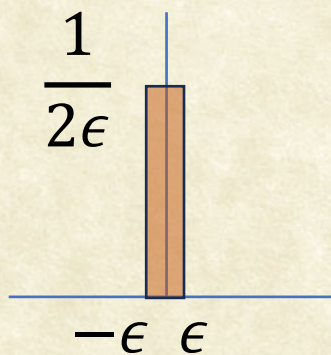
Monsoon 2025: I: Impulse Response



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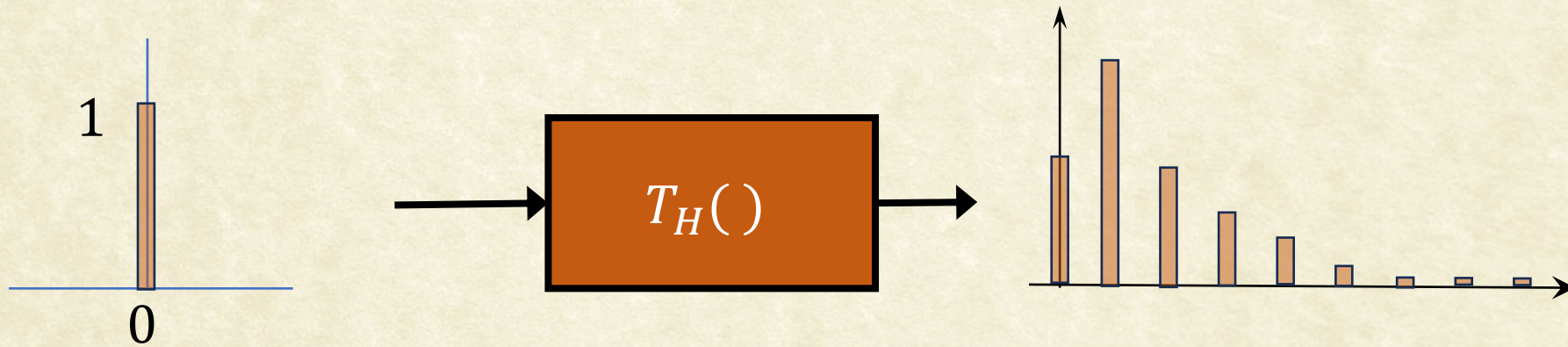
Impulse Response





Impulse Response: Discrete Signals

- Input is unit impulse: 1 at $t=0$; 0 elsewhere

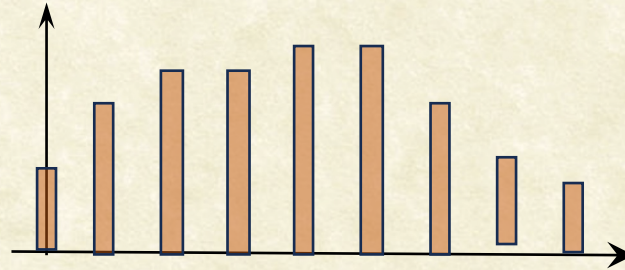


- Qn: Can the response start before 0?

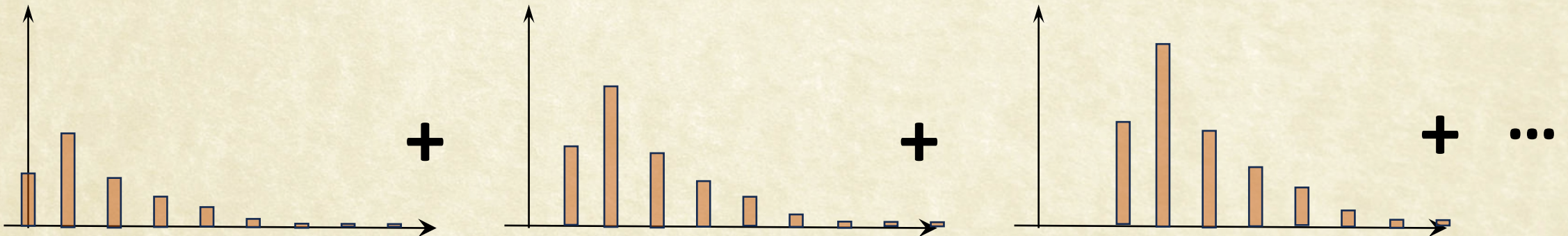


What about Arbitrary Signals?

- Any signal can be thought of as a sum of **scaled** and **shifted** impulse functions



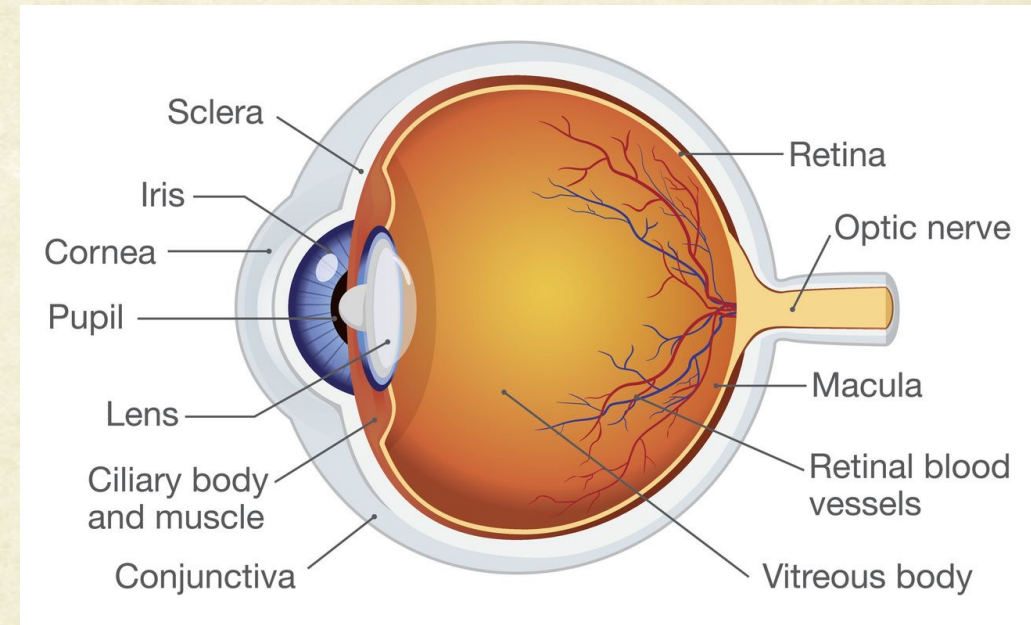
- If the system is linear and shift invariant,
 - The result will be the sum of **scaled** and **shifted** impulse responses





Impulse Response of Human Eye

- Input: Point Light Source ?
 - A bright (distant) star
- System: Cornea, Eye Lens, Vitreous, Retina
- Output: PSF



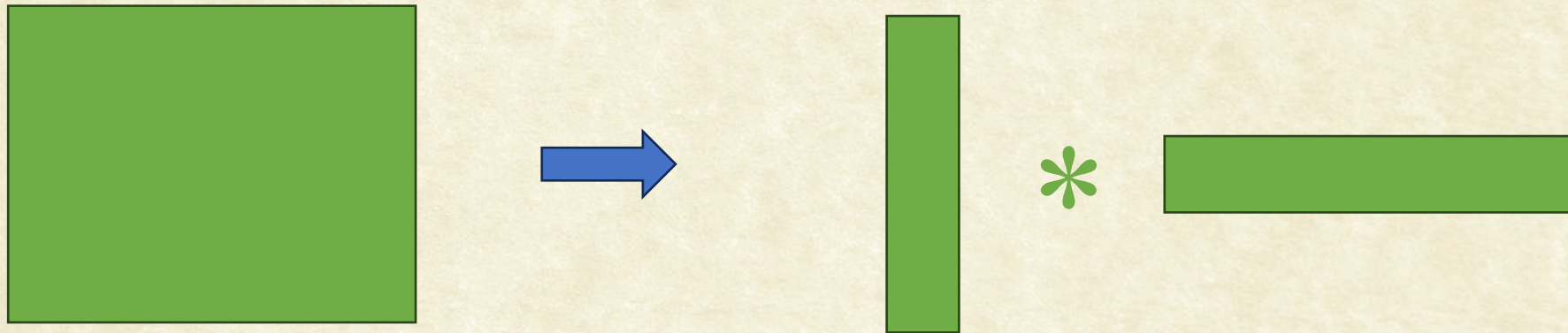


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: **II: Separability**



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Separable Kernels

- A 2D function is separable if it can be written as the product of two 1D functions:

$$G(x, y) = G_1(x)G_2(y).$$

- A separable kernel can be written as the outer product of two vectors:

$$w = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = er^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} [1 \quad 1 \quad 1]$$

- Convolution with a separable 2D kernel can be implemented as two 1D kernels
 - Convolution of an $M \times N$ image with an $m \times n$ kernel takes $MNmn$ multiplications and additions.
 - How much does two 1D convolutions cost?



Questions

- Show that 2D Gaussian kernel is separable [Homework]
- What is the rank of a separable kernel?
- How do we find the 1D components of a kernel (provided it is separable)?

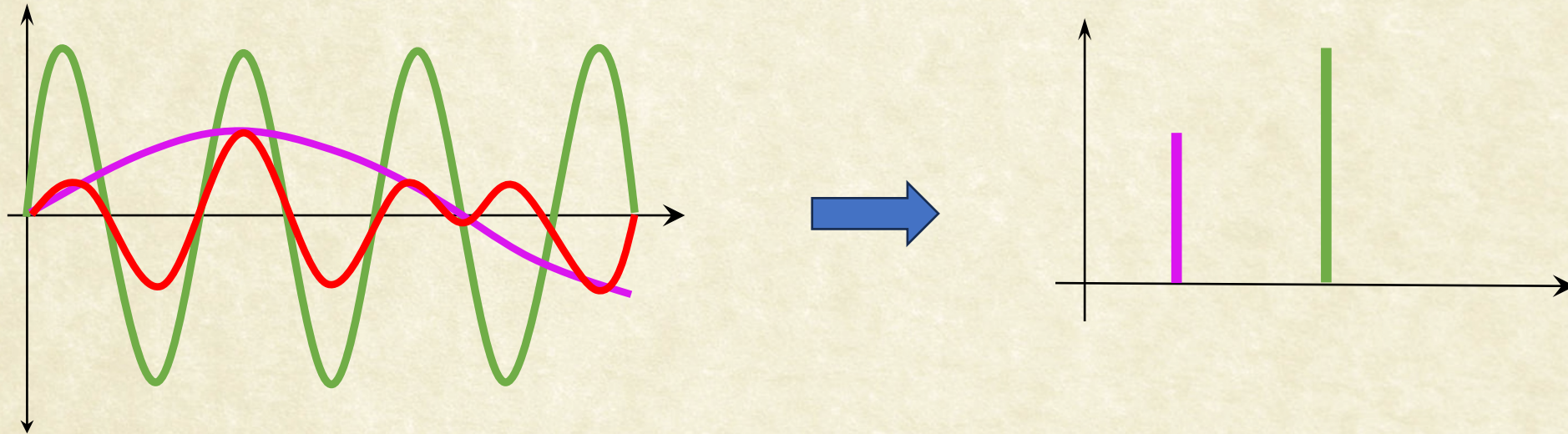


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: **III**: Frequency Domain Basics

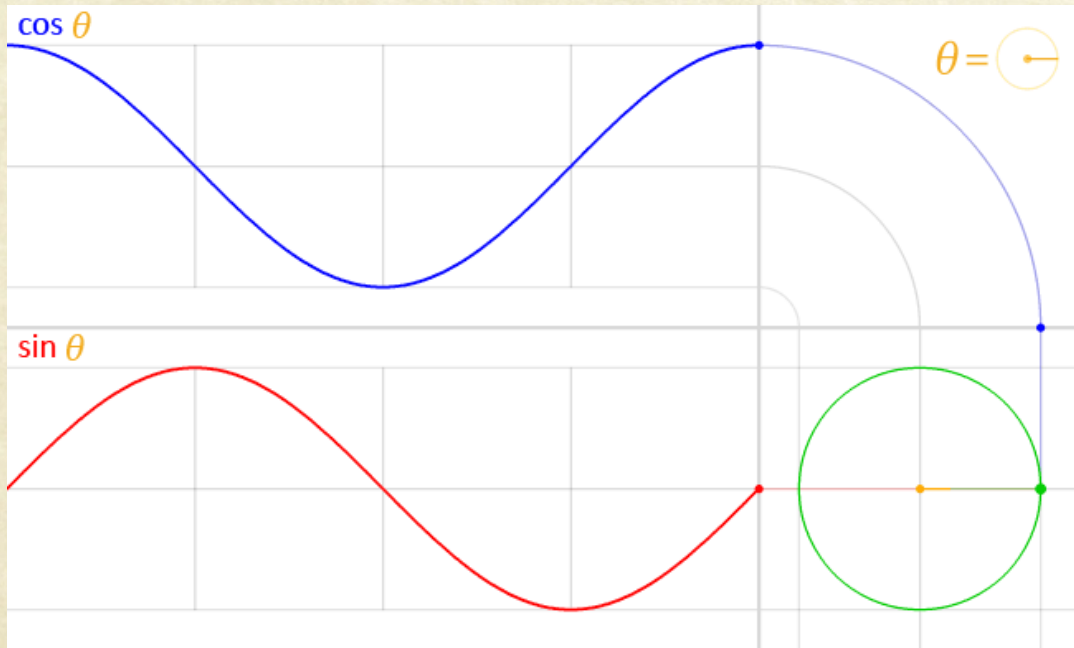


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Periodic Signals

- **Periodic** → **Frequency** of occurrence
 - Repetitions/<Unit> (**cycles**/sec = Hz)



$$\frac{4 \sin \theta}{\pi}$$


$$\frac{4 \sin 3\theta}{3\pi}$$


$$\frac{4 \sin 5\theta}{5\pi}$$


$$\frac{4 \sin 7\theta}{7\pi}$$


$$x(t) = A \cos(\omega t) = A \cos(2\pi f t) = A \cos\left(\frac{2\pi}{T} t\right)$$

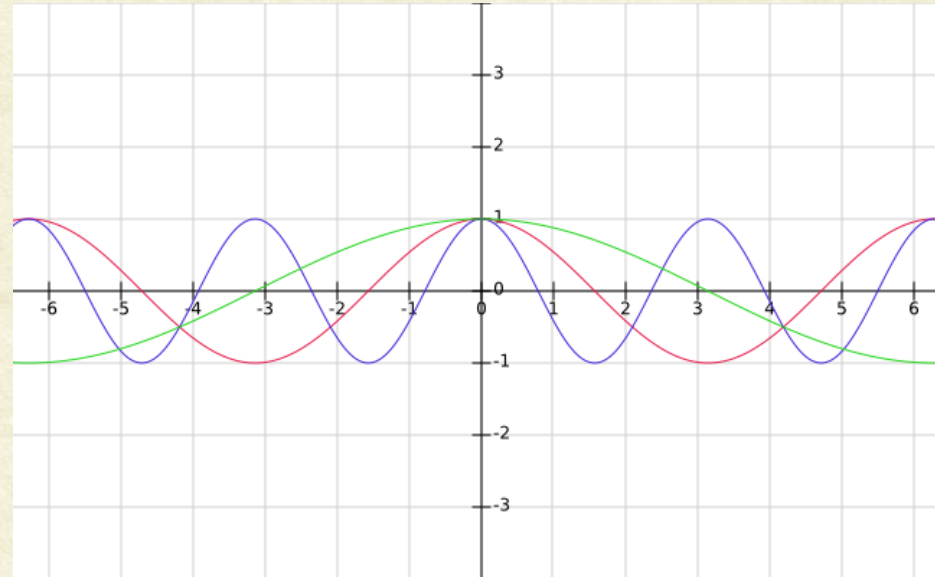
Angular frequency

Fundamental Period



Simple periodic signals

- $x(t) = A \cos(t)$
- $x(t) = A \cos(2t)$
- $x(t) = A \cos(t/2)$



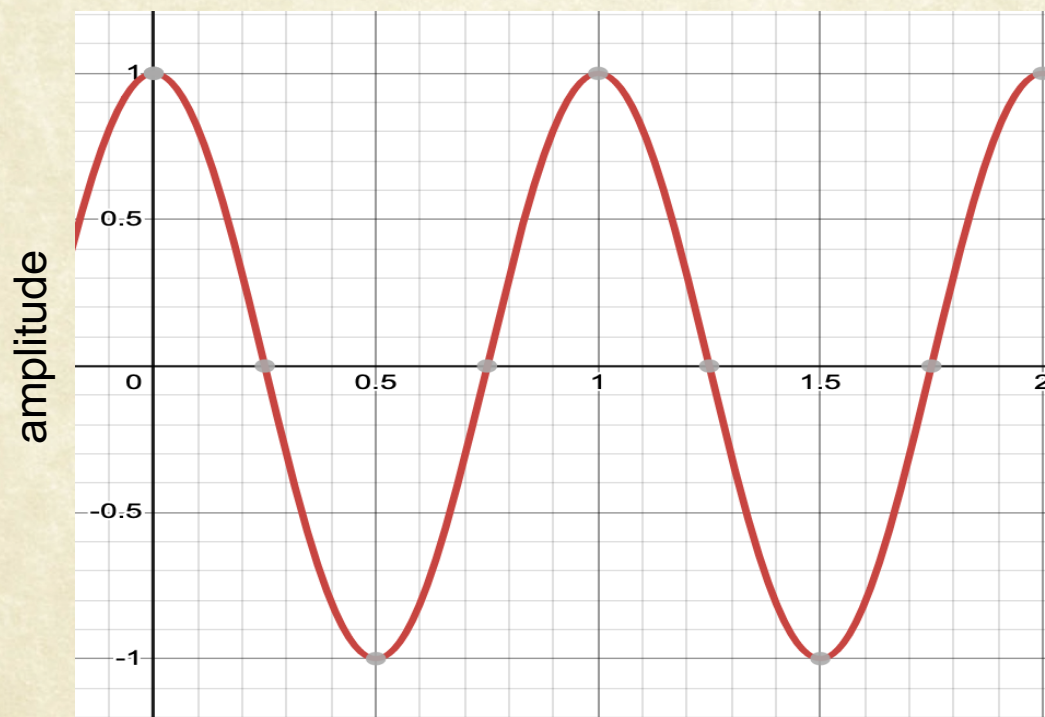
- $x(t) = A \cos(\omega t) = A \cos(2\pi f t) = A \cos(\frac{2\pi}{T} t)$

Angular frequency



Signal and Frequency Domains

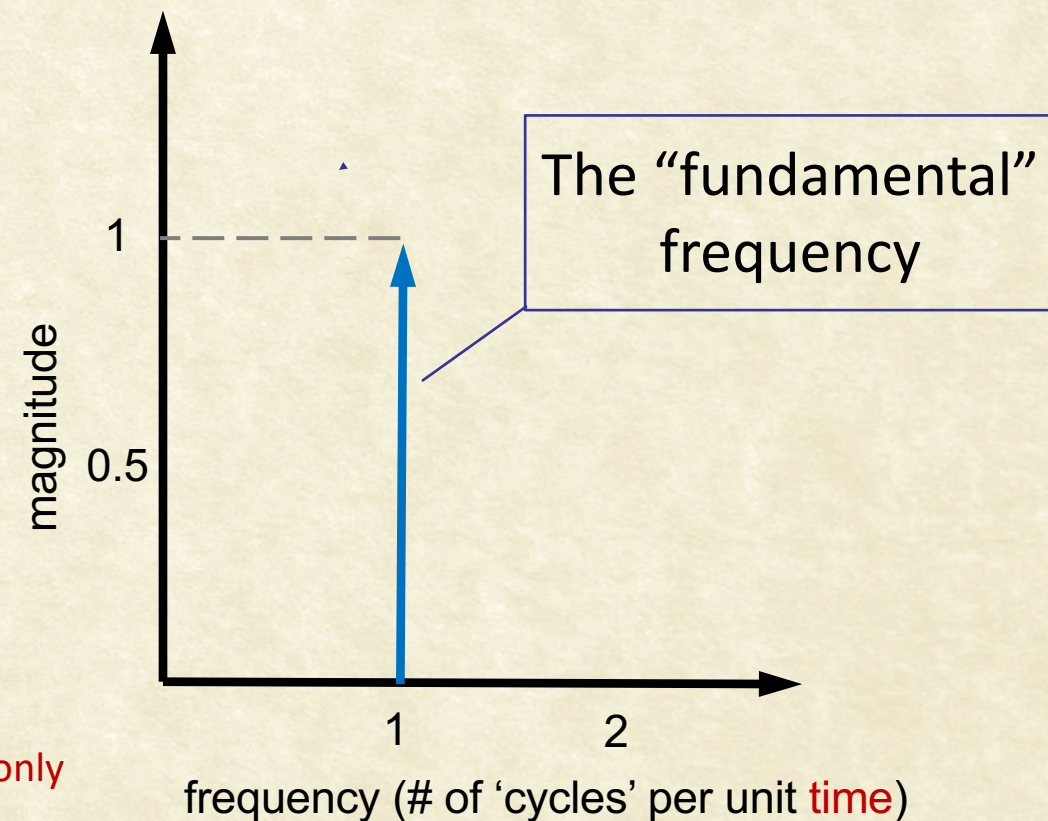
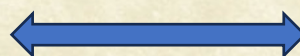
$$y = A \cos(2\pi ft) = \cos(2\pi t)$$



NOTE: $t=0$ is time from where signal is considered

NOTE: t is bounded only for physical signals

Time domain

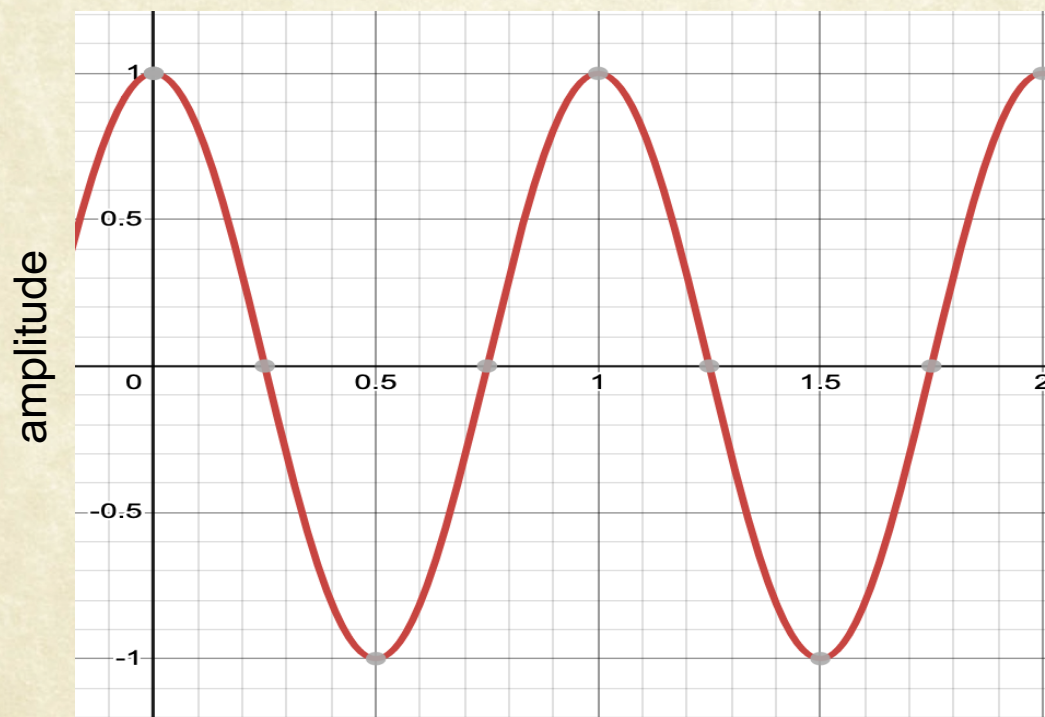


Frequency domain



Signal and Frequency Domains

$$y = A \cos(2\pi f s) = \cos(2\pi s)$$

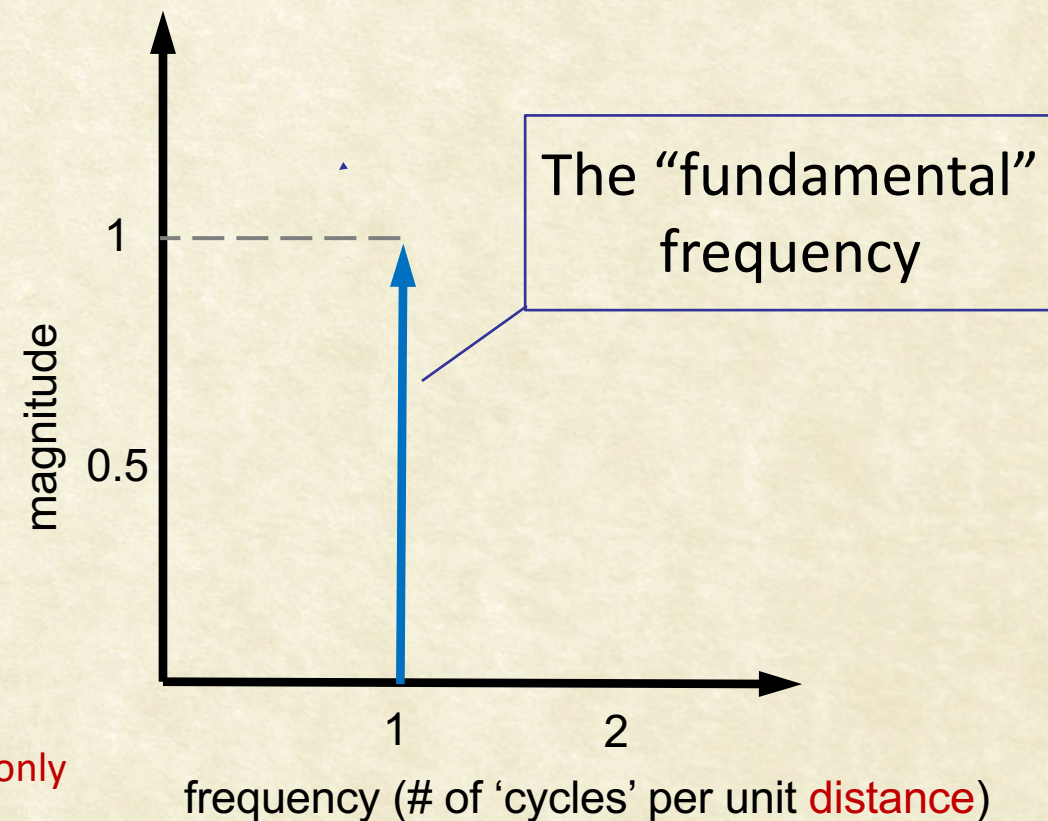
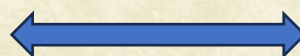


NOTE: $s=0$ is time from where signal is considered

Spatial distance (**s**)

NOTE: s is bounded only for physical signals

Spatial domain

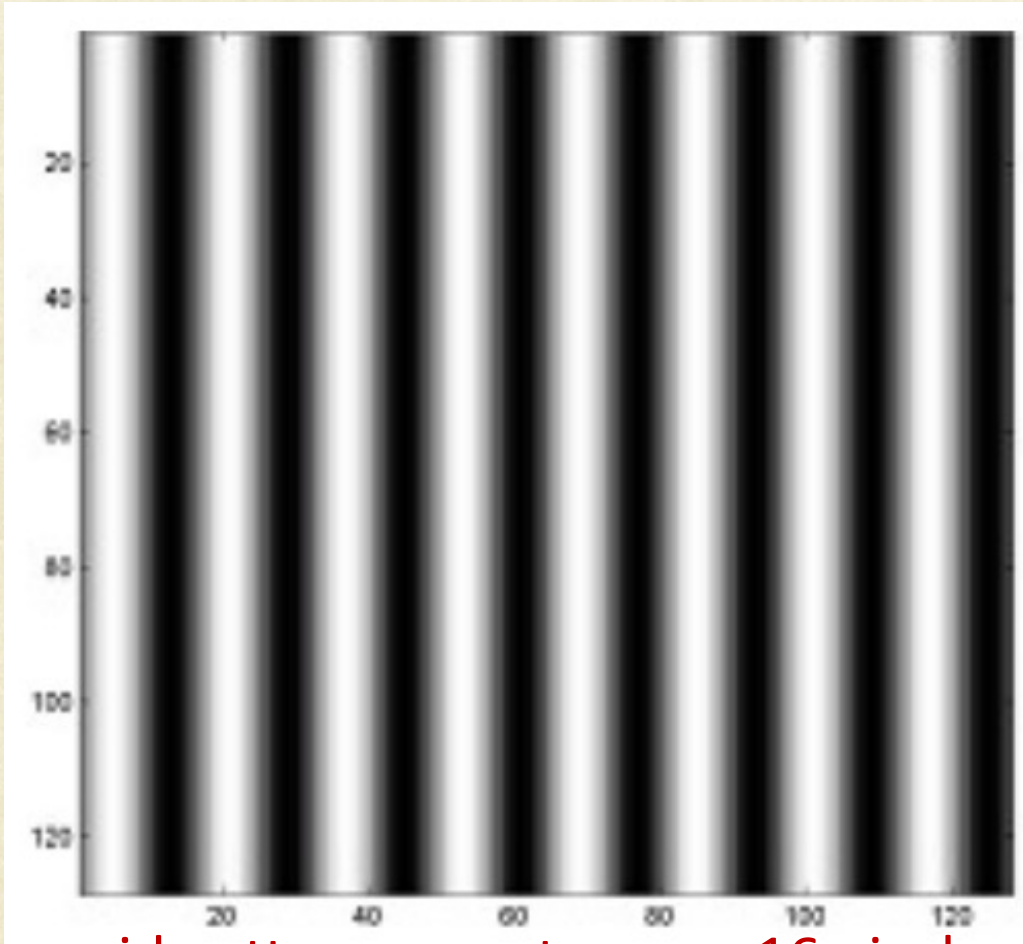


Frequency domain



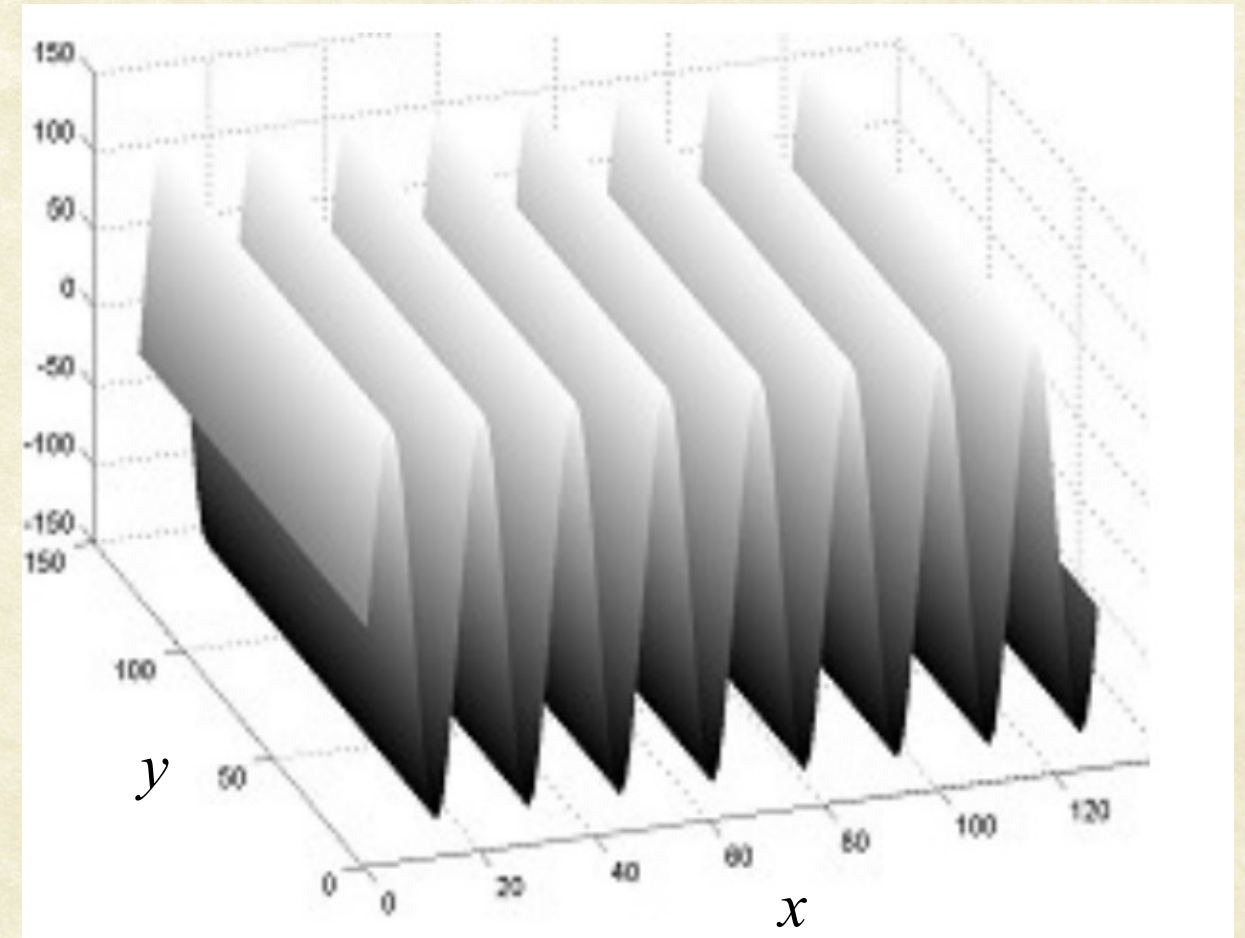
Periodic Signals in Images

128 x 128 grayscale image



Sinusoid pattern repeats every 16 pixels
 $f = 1/16$ cycles/pixel

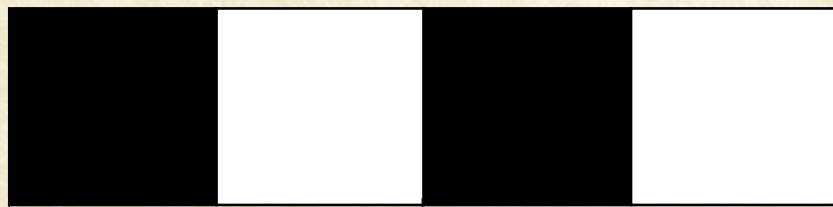
$$I(x, y) = 128 \sin(2\pi x/16)$$





Periodic Signals in Images

- Spatial period = Minimal # of pixels between two identical patterns in a “periodic” image



Minimal Period:
2 pixels

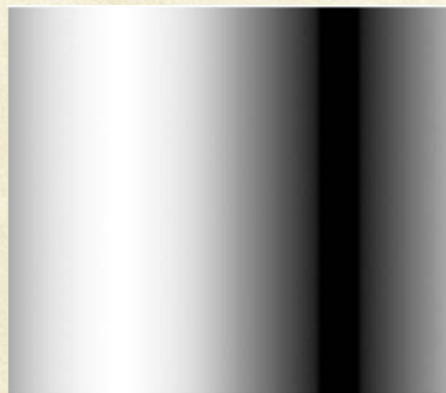


$$v_{max} = \frac{1}{\text{minimal period}} = \frac{1}{2}$$



Extending sinusoids to 2D

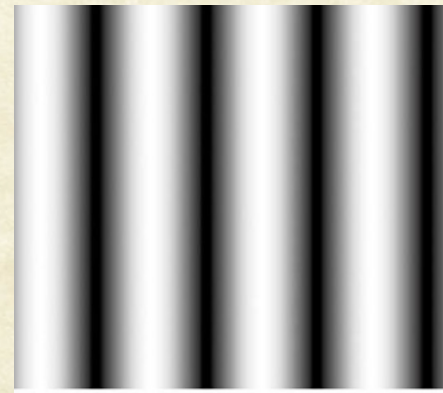
$$s(x, y) = \sin[2\pi(u_0x + v_0y)]$$



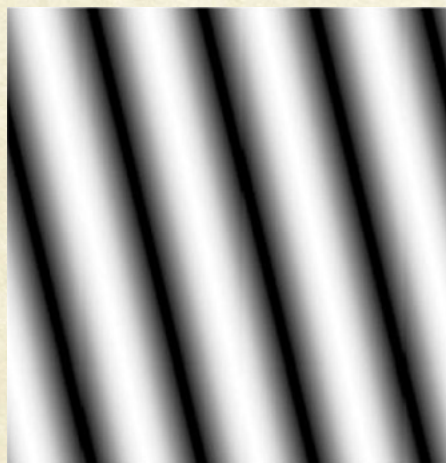
$$u_0 = 1, v_0 = 0$$



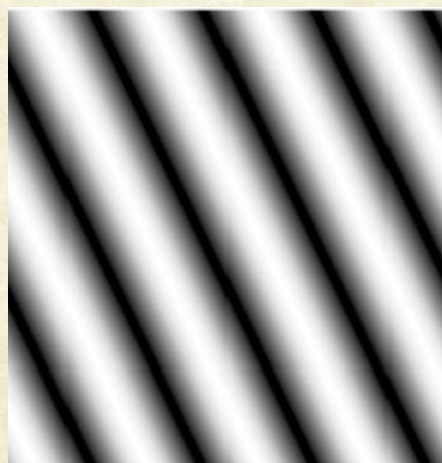
$$u_0 = 2, v_0 = 0$$



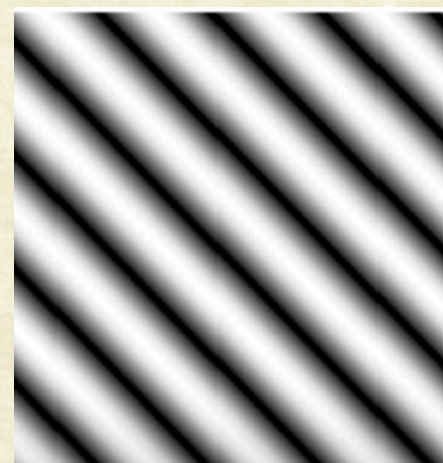
$$u_0 = 4, v_0 = 0$$



$$u_0 = 4, v_0 = 1$$



$$u_0 = 4, v_0 = 2$$

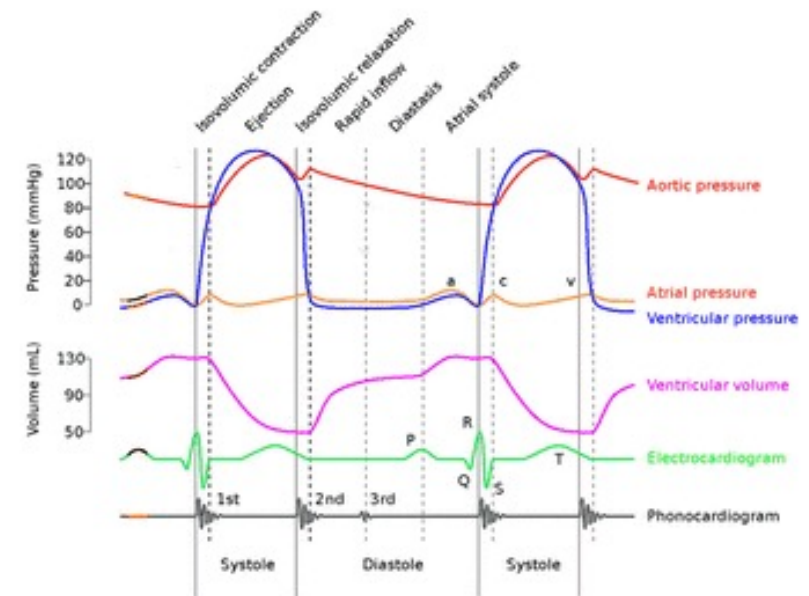
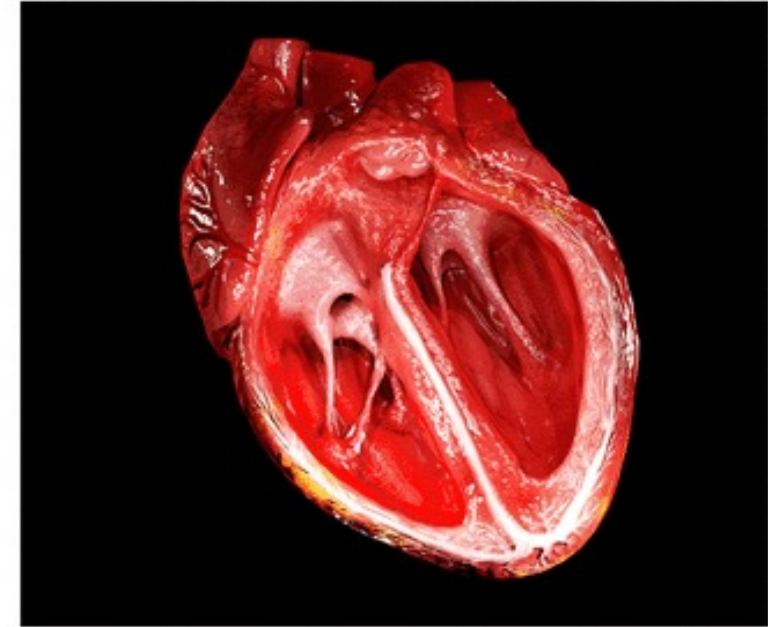


$$u_0 = 4, v_0 = 4$$



Many natural phenomena (signals) are periodic but not necessarily sinusoidal

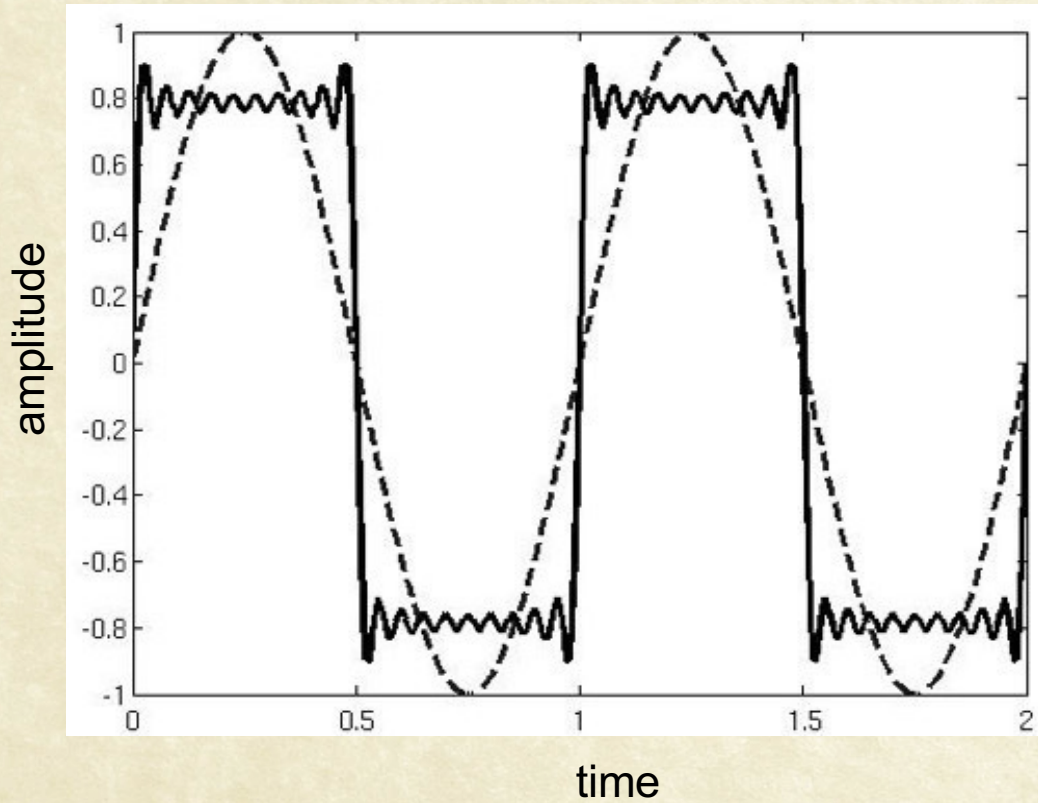
CARDIAC CYCLE



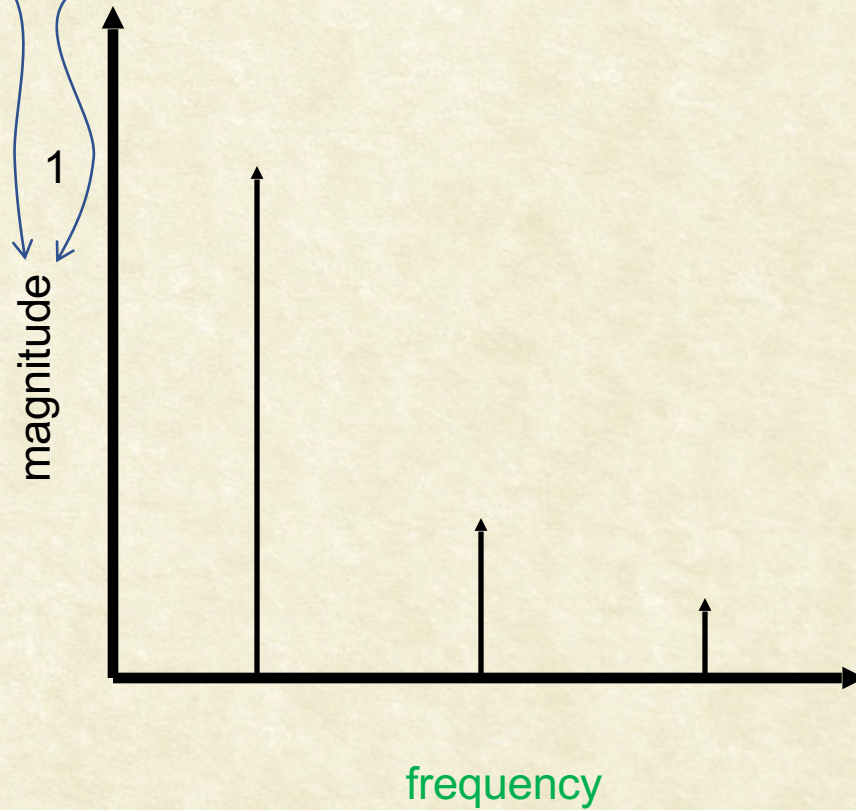


Fourier Series

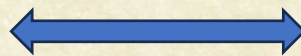
$$y(t) = \sum_n a_n \sin(nf \times 2\pi t) + \sum_n b_n \cos(nf \times 2\pi t)$$



Signal domain



Frequency Spectrum



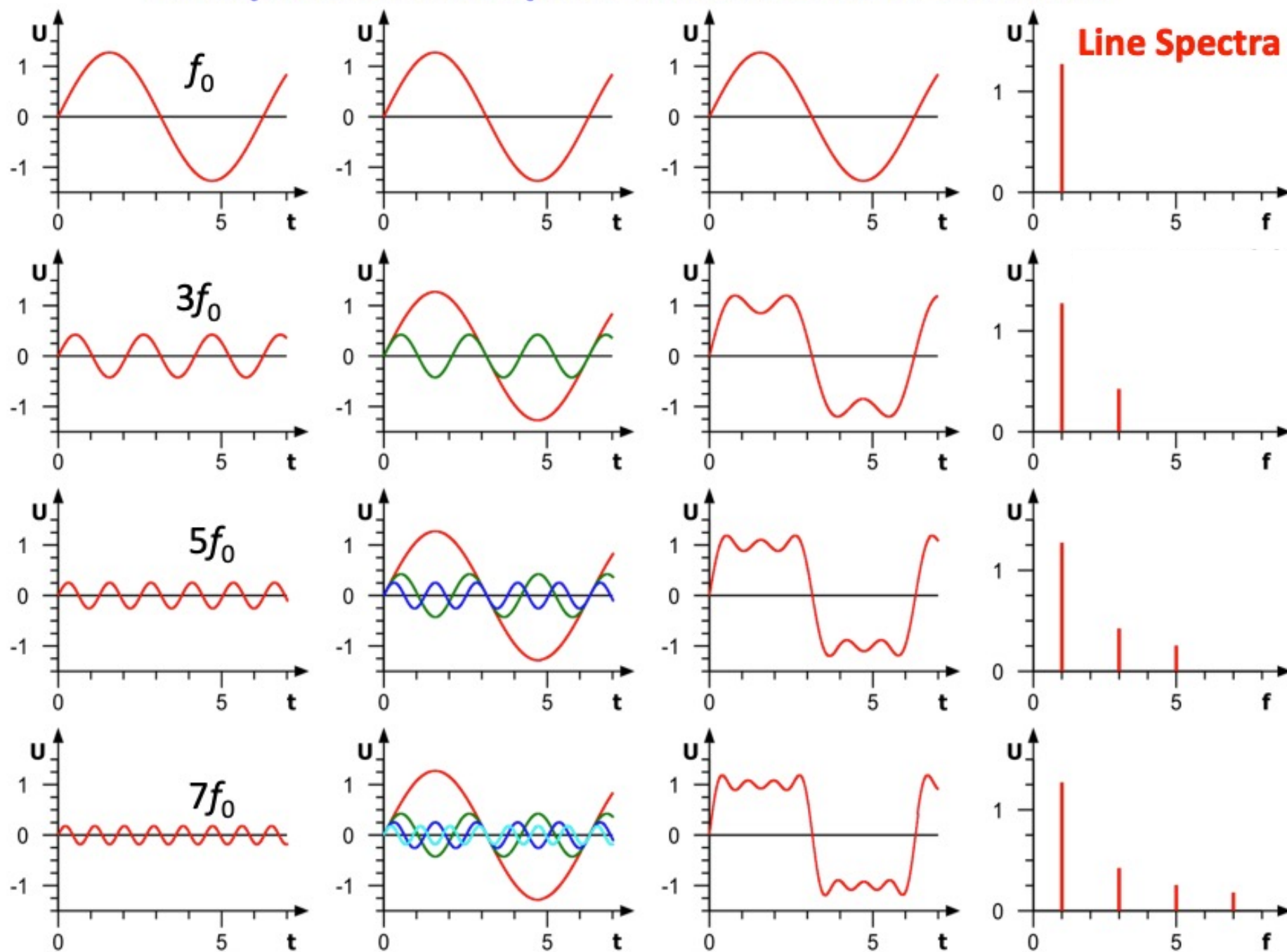


Fourier Series, visually



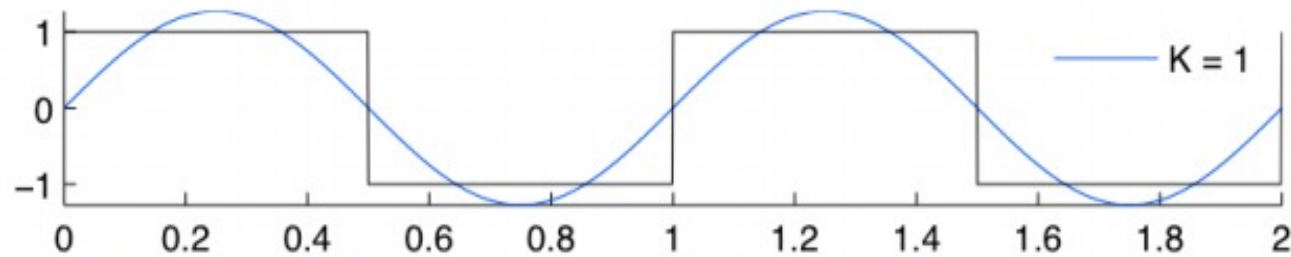


Example: Periodic Square Wave as Sum of Sinusoids

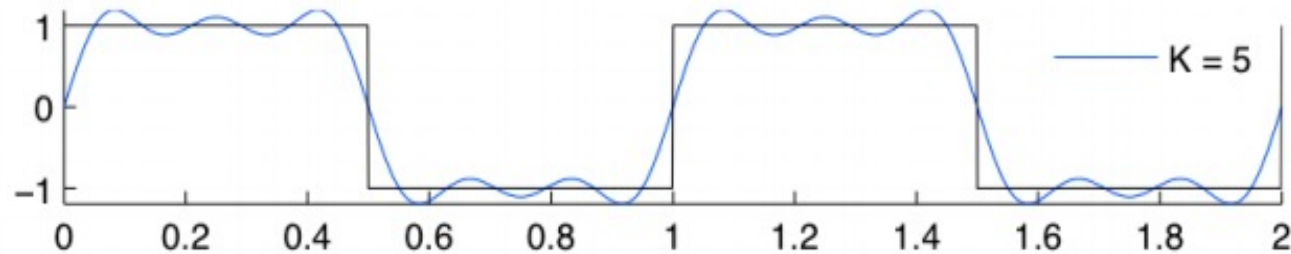




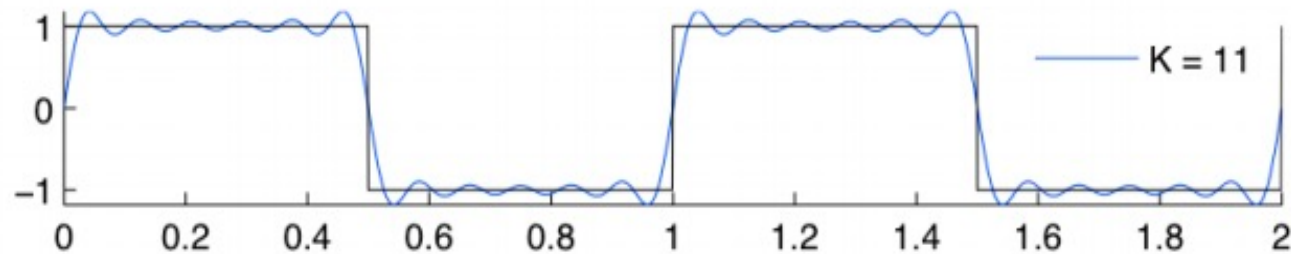
$$f(t) = \frac{4}{\pi} \left[\sin(\pi t) + \frac{1}{3} \sin(3\pi t) + \frac{1}{5} \sin(5\pi t) + \frac{1}{7} \sin(7\pi t) + \dots \right]$$



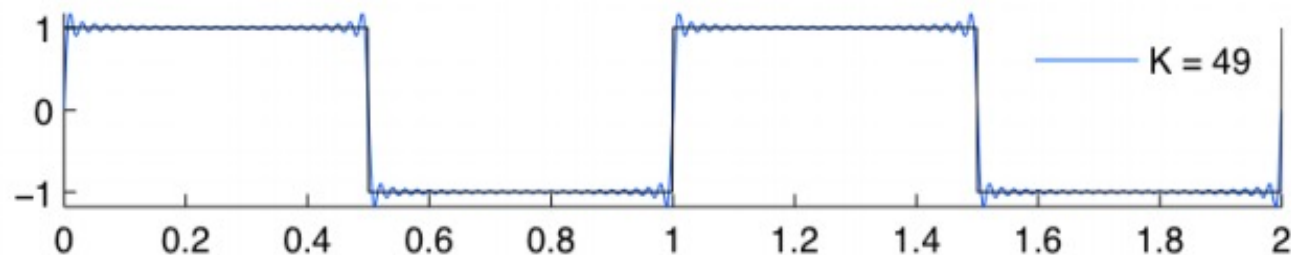
Fundamental only



Five terms



Eleven terms



Forty-nine terms

$t \rightarrow$



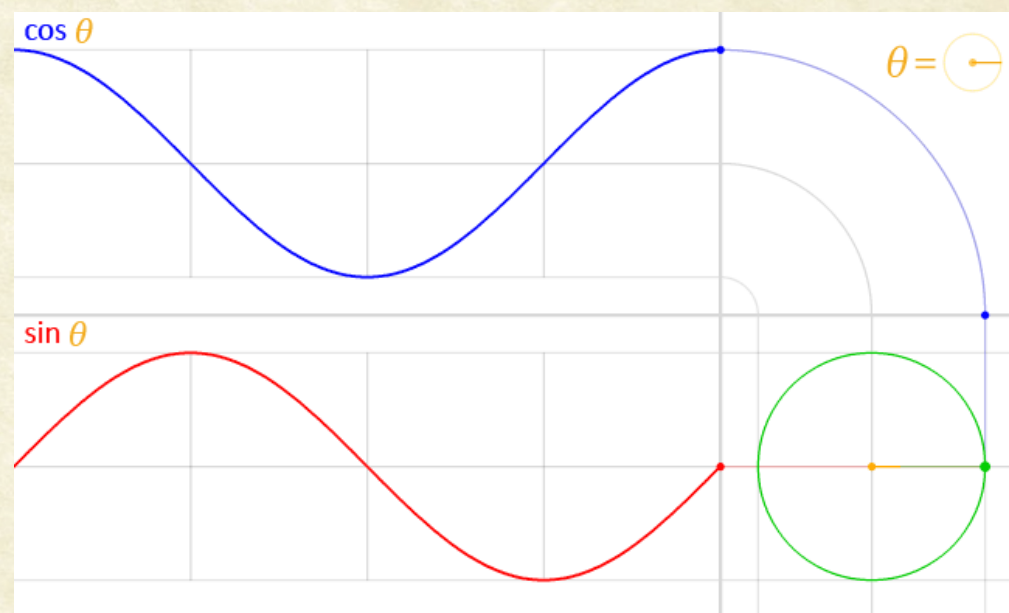
be rational
 i
guys...
 π
get real

$$e^{i\pi} + 1 = 0$$

Euler's identity:
uniting constants
since 1748

$$e^{it} = \cos t + i \sin t$$

$$i = \sqrt{-1}$$





Complex sinusoid

$$e^{it} = \cos t + i \sin t$$

$$i = \sqrt{-1}$$

